

Shahzaib Naseem

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PROFESSIONAL SUMMARY

Final-year Computer Science student (AI Specialization) at University of Education, Lahore, with hands-on experience building production-ready AI systems spanning large language models, generative AI, NLP pipelines, and deep learning. Developed and deployed LLM-powered applications including RAG-based chatbots, n8n AI automation agents, an AI Interview Coach, HR Assistant, AI PDF Reader, and Customer Support Agents. Proficient in Python, PyTorch, TensorFlow, LangChain, HuggingFace Transformers, OpenAI API, and Flask/FastAPI. Seeking an AI/ML Engineering internship to contribute scalable, real-world intelligent systems.

TECHNICAL SKILLS

Generative AI / LLMs	OpenAI API (GPT-4/3.5), LangChain, HuggingFace Transformers, Prompt Engineering, RAG Pipelines, LLM Fine-Tuning (LoRA), Embeddings, Semantic Search
ML / Deep Learning	Python, PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LSTM, Bidirectional LSTM, Attention Mechanism, Transfer Learning, CNNs
NLP	Tokenization, Lemmatization, MFCC Extraction, Named Entity Recognition, Text Classification, Sentiment Analysis, HuggingFace Datasets
Vector Databases	FAISS, ChromaDB, Pinecone (fundamentals), Cosine Similarity Search, Document Chunking
AI Automation	n8n Workflow Automation, AI Agents, Tool-Use Pipelines, Webhook Integrations, Personal Assistant Agents, Customer Support Automation
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Data Augmentation, GridSearchCV, Experiment Tracking
Web & Deployment	Flask REST APIs, FastAPI, Streamlit, React.js, Node.js, Express.js, MongoDB, JWT Auth, Docker (fundamentals)
Tools & Platforms	Git, GitHub, VS Code, Jupyter Notebook, Postman, SQL, HuggingFace Hub, Google Colab

PROJECTS

Generative AI & LLM Applications

AI Interview Coach

2024

Python · LangChain · OpenAI GPT-4 · HuggingFace Transformers · FAISS · FastAPI · Streamlit

- Built an end-to-end AI-powered interview coaching chatbot using LangChain and GPT-4, supporting role-specific Q&A; generation for AI/ML, Data Science, and software engineering positions.
- Implemented RAG pipeline with FAISS vector store and OpenAI Embeddings to retrieve relevant interview questions and expert answers from a curated knowledge base of 500+ Q&A; pairs.
- Deployed interactive Streamlit web app with real-time feedback generation; reduced interview preparation time by an estimated 60% compared to manual study.

HR Assistant Chatbot

2024

Python · LangChain · OpenAI API · ChromaDB · FastAPI · React.js · MongoDB

- Developed an LLM-powered HR assistant chatbot capable of answering policy queries, onboarding FAQs, and leave management questions using a RAG architecture over internal HR documents.
- Built document ingestion pipeline supporting PDF, DOCX, and TXT formats with semantic chunking and ChromaDB vector storage, achieving sub-2-second query response time.
- Integrated FastAPI backend with React.js frontend and JWT authentication; system handled 200+ simulated concurrent HR queries with consistent accuracy across intent categories.

AI PDF Reader & Document Intelligence

2024

Python · LangChain · OpenAI Embeddings · FAISS · Streamlit · PyPDF2

- Created an intelligent PDF Q&A; system using LangChain's document loaders, recursive text splitter for optimal chunking, and FAISS for fast similarity retrieval over uploaded documents.
- Implemented conversational memory to maintain multi-turn dialogue context, enabling follow-up questions across 50+ page documents with accurate source citation.
- Deployed as a Streamlit application supporting multi-PDF upload; reduced document research time by approximately 70% compared to manual reading in user testing.

AI Automation (n8n)

Personal AI Assistant — n8n Automation Agent

2024

n8n · OpenAI API · Webhook Integration · Google Sheets · Gmail · Telegram Bot API

- Designed and deployed an autonomous personal assistant workflow in n8n integrating OpenAI GPT-4 for natural language understanding, automating task scheduling, email drafting, and reminder management.
- Built multi-step agentic pipeline with conditional branching and tool-use nodes (calendar, email, search), reducing manual task management overhead by approximately 80%.
- Connected Telegram Bot API as the user interface, enabling voice-to-text transcription and natural language commands for workflow triggering.

Customer Support AI Agent — n8n Pipeline

2024

n8n · OpenAI API · LangChain · Webhook · MongoDB · Slack Integration

- Built a fully automated customer support agent using n8n orchestration with OpenAI-powered intent classification, routing queries to appropriate resolution paths without human intervention.
- Implemented escalation logic, FAQ retrieval from MongoDB knowledge base, and Slack notification for unresolved queries; achieved 85% automated resolution rate in testing.
- Reduced simulated first-response time from hours to under 30 seconds across 15+ support intent categories.

Machine Learning & Deep Learning

Speech Emotion Recognition System

2024

Python · TensorFlow · Bidirectional LSTM · Attention Mechanism · MFCC · Flask · React.js

- Built end-to-end audio emotion classification pipeline on the RAVDESS dataset using MFCC feature extraction with data augmentation (noise injection, time stretching), achieving 84% test accuracy across 6 emotion classes.
- Designed BiLSTM + Self-Attention architecture that outperformed vanilla LSTM baseline by 11% F1-score, with dropout and L2 regularisation reducing overfitting on unseen speakers.
- Deployed as Flask REST API with React.js frontend; achieved sub-second real-time inference and automated approximately 60% of the manual audio preprocessing pipeline.

Reusable ML Pipeline — Classification & Prediction Framework

2024

Python · Scikit-learn · XGBoost · Pandas · Matplotlib · Seaborn · MLflow

- Designed modular, dataset-agnostic ML pipeline (ingestion → EDA → cleaning → feature engineering → model training → evaluation → reporting), reducing new dataset setup time by 50%.
- Benchmarked Logistic Regression, Random Forest, and XGBoost with GridSearchCV hyperparameter tuning; XGBoost achieved best performance (F1: 0.91, ROC-AUC: 0.96) on a 50K-sample classification dataset.
- Integrated MLflow experiment tracking for reproducible runs; documented all experiments with parameter logs, metric comparisons, and model artefact storage.

EDUCATION

B.Sc. Computer Science — AI Specialization

Sep 2022 – Oct 2026

University of Education, Lahore

Relevant Coursework: Machine Learning · Deep Learning · NLP · Generative AI · Data Structures & Algorithms · Database Systems · Computer Vision

CERTIFICATIONS

Machine Learning Specialization — Coursera — DeepLearning.AI (Andrew Ng)

2024

Advanced Web Development — MERN Stack — Coursera

2024

ADDITIONAL

Languages: English (Professional) · Urdu (Native)

Interests: Generative AI research, LLM applications, NLP, AI automation, open-source ML, building AI-powered SaaS products for Pakistan's tech ecosystem